

Anaphylaxis to pork kidney is related to IgE antibodies specific for galactose- α -1,3-galactose.



BACKGROUND: Carbohydrate-specific IgE antibodies present on nonprimate mammalian proteins were incriminated recently in delayed meat anaphylaxis. The aim of this study was to explore whether anaphylaxis to mammalian kidney is also associated with galactose- α -1,3-galactose (α Gal)-specific IgE. **METHODS:** Fourteen patients with anaphylaxis to pork or beef kidney underwent prick tests to meat and kidney. Some patients also underwent skin tests to Erbitux(®) (cetuximab). IgE antibodies to α Gal, swine urine proteins, beef and pork meat, serum albumin proteins, cat, and rFel d 1 were measured by ImmunoCAP(®). The α Gal levels were estimated in meats and kidney by ELISA inhibition assay. Cross-reactivity between α Gal and pork kidney was studied with the ImmunoCAP(®) inhibition assay. **RESULTS:** Among the 14 patients, 12 presented with anaphylactic shock. Reactions occurred within 2 h from exposure in 67% of patients. Associated risk factors were observed in 10 cases, and alcohol was the main cofactor. Three patients underwent an oral challenge to pork kidney, and anaphylaxis occurred after ingestion of small quantities (1-2 g). Prick tests to kidney were positive in 54% of patients. All tested patients showed positive skin tests to Erbitux(®). All patients tested positive for IgE to α Gal, with levels ranging from 0.4 to 294 kU/l. IgE binding to α Gal was inhibited by raw pork kidney extract (mean, 77%; range, 55-87%), which showed a high amount of α Gal determinants. **CONCLUSIONS:** Pork or beef kidney anaphylaxis is related to α Gal IgE. Its peculiar severity could be due to an elevated content of α Gal epitopes in kidney. © 2012 John Wiley & Sons A/S.